ATUL RAJ

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PROFESSIONAL SUMMARY

B.Tech graduate (EEE, VIT Vellore, May 2026) with 3 production-deployed full-stack applications and measurable engineering outcomes: 91% ML classification accuracy, sub-200ms API response times, and 60% database load reduction under concurrent-user stress tests. Proven ability to collaborate in team environments, mentor peers, and manage large-scale operations — aligned with Philips' mission of human-centred technology and people-first culture. Eager to contribute as a Software Developer Apprentice Trainee in health-technology innovation.

TECHNICAL SKILLS

Languages: Python, SQL, C, HTML5, CSS3
Frameworks: Django, Django REST Framework, FastAPI, Flask, Bootstrap, React.js
ML / Data: scikit-learn, NumPy, pandas, Plotly, TF-IDF, Logistic Regression
Databases & DevOps: PostgreSQL, Redis, SQLite — Docker, AWS (EC2, S3, Lambda), Git, Heroku, Gunicorn, Nginx
Concepts: RESTful APIs, ORM, JWT Auth, OAuth, Agile/Scrum, Unit Testing (Pytest), CI/CD
Soft Skills: Cross-functional collaboration, technical mentoring, stakeholder communication, event operations

EXPERIENCE

Embedded Software Intern | Maven Silicon *May 2025 – Jul 2025 • Remote, India*

- Developed and optimised **5+ C firmware modules** for ARM/RISC-V targets under strict 64 KB heap constraints, reducing critical-path execution time by **~20%** — demonstrating ability to write reliable, resource-efficient software under real engineering constraints.
- Managed Git version control for a 4-engineer team, enforcing branching conventions that cut merge conflicts by **30%**; resolved **12+ bugs** before system-level validation — reflecting the collaborative, quality-first mindset valued at Philips.

PROJECTS

Personal Finance Dashboard *| Django, PostgreSQL, React.js, Plotly, AWS*

- Architected **12 RESTful APIs** (JWT-secured) processing 1,000+ transaction records at **sub-200 ms** via PostgreSQL indexing; integrated 5 Plotly visualisations for real-time spending insights — end-to-end full-stack ownership from backend to UI.
- Deployed via Gunicorn + Nginx on AWS/Heroku with automated DB backups; achieved **99.5% uptime** across 3 months in production — demonstrating deployment discipline and service reliability.

Fake News Detection System *| FastAPI, scikit-learn, React.js, PostgreSQL, Docker*

- Trained TF-IDF + Logistic Regression classifier on 20K articles achieving **91% accuracy**; served predictions via FastAPI with end-to-end latency under **1.2 s** — built for real-world usability, not just experimentation.
- Reduced SQL retrieval time by **40%** via composite indexing; Dockerised full stack (FastAPI + React + PostgreSQL) for one-command deployment, simplifying onboarding for any collaborator.

Chirper — Micro-Blogging Platform *| Django, Bootstrap, PostgreSQL, AWS S3, Redis*

- Implemented full authentication (session + OAuth), CRUD operations, and follow-graph using Django ORM; offloaded media to AWS S3, cutting image load time by **35%**.
- Resolved N+1 queries with select_related/prefetch_related; Redis feed caching cut DB hits by **60%**, sustaining **50+ concurrent users** under Locust stress tests — production-grade scalability from the ground up.

EDUCATION

Vellore Institute of Technology *Vellore, Tamil Nadu • Sept 2022 – May 2026*

B.Tech — Electrical and Electronics Engineering — CGPA: 7.54 / 10

- Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

CERTIFICATIONS & LEADERSHIP

Certifications: Microsoft — Intro to Generative AI | NVIDIA — Accelerated Computing with CUDA Python | Maven Silicon — Embedded Systems Design

Senior Core Member, ML Domain | Apple Developer Student Club — VIT *2022 – 2026 • Vellore*

- Directed 'iOS Fusion 6.0,' a large-scale technical event at Gravitas fest, managing end-to-end operations for **1,500+ participants** — demonstrating project management and cross-team coordination at scale.
- Conducted technical workshops and mentored **50+ junior students** in Python and ML; built a culture of learning — aligned with Philips' emphasis on knowledge-sharing and people development.

Operations & Sales Head | Gravitas & Riviera — VIT National Festivals *2023 – 2025 • Vellore*

– Led sponsor outreach, pitch presentations, and sales operations for VIT's national technical and cultural festivals across **2 consecutive years**; managed a team of **20+ juniors**.